Quantum Computing

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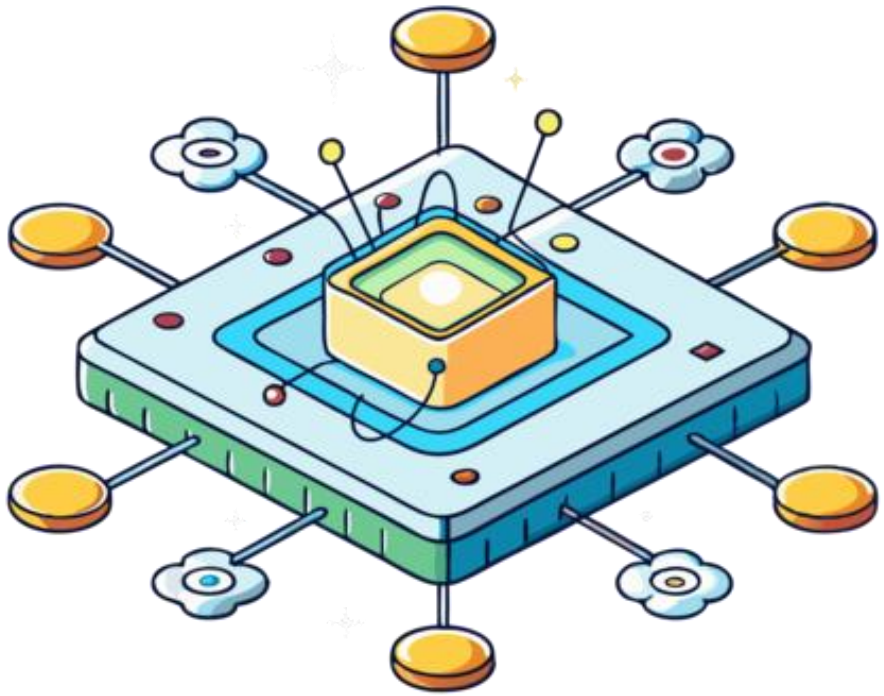
Learning Objective

- Quantum computing-Definition
- Practical applications for quantum computing
- Four key principles of quantum mechanics
- How do quantum computers work?
- Quantum computing components
- Classical computing versus quantum computing
- When is quantum computing superior?
- Quantum computing use cases
- Quantum utility versus quantum advantage
- Quantum benchmarks
- Quantum challenges and Improvement

Quantum computing

Quantum computing is an emergent field of computer science and engineering that harnesses the unique qualities of quantum mechanics to solve problems beyond the ability of even the most powerful classical computers.

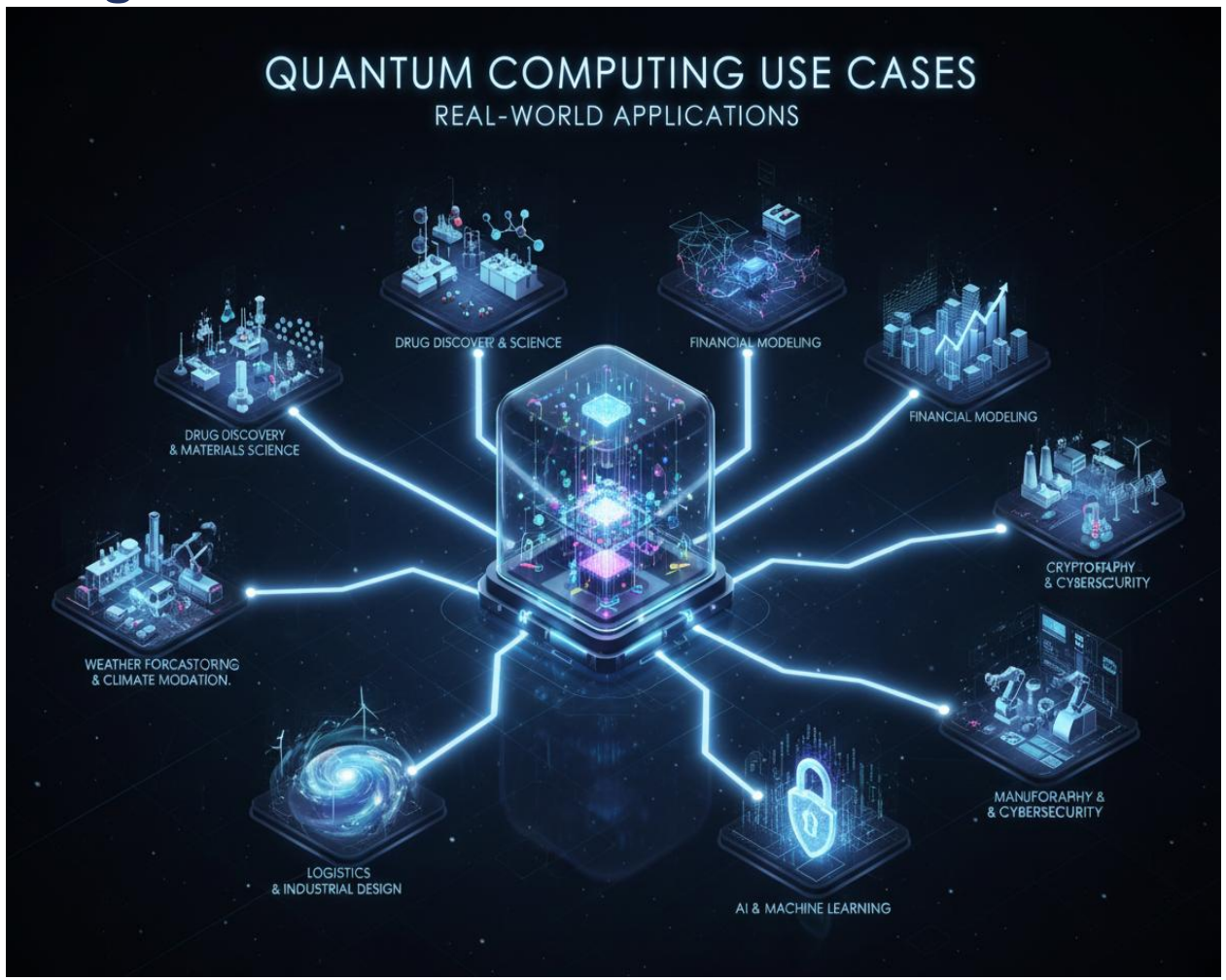
- Quantum computing is a type of computing based on quantum mechanics
- It uses qubits instead of classical bits
- It can solve complex problems beyond classical computers
- Works alongside classical computing



Real-time Applications For Quantum Computing

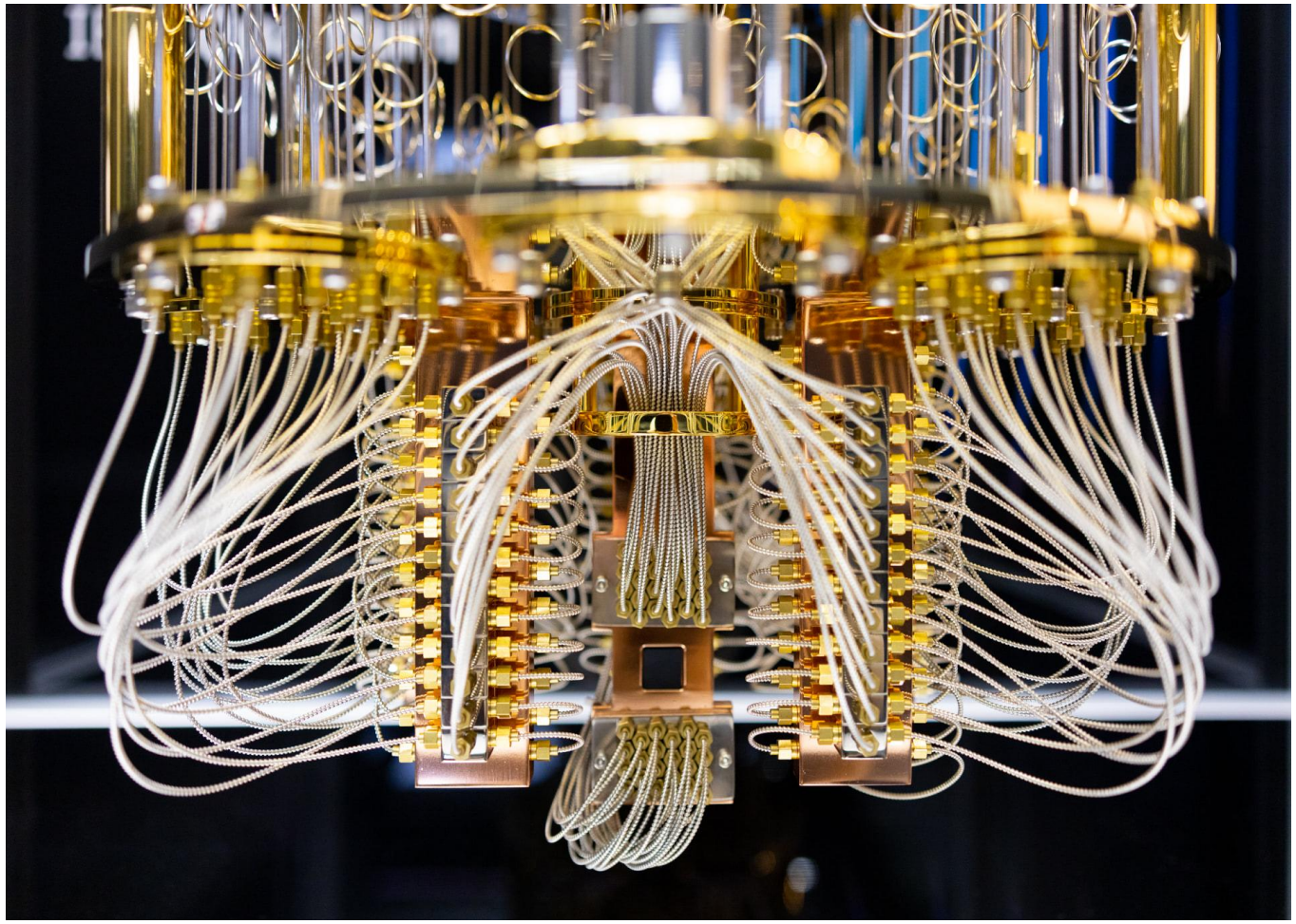
- In practice, quantum computers are expected to be broadly useful for two types of tasks: modeling the behavior of physical systems and identifying patterns and structures in information.

- **Drug discovery**
  - Helps design new medicines
  - Simulates molecular structures
  - Speeds up research process
- **Materials science**
  - Helps discover new materials
  - Improves product design
  - Used in energy and manufacturing
- **Financial services**
  - Improves risk analysis
  - Helps optimize investment strategies
  - Detects fraud patterns



Four key principles of quantum mechanics

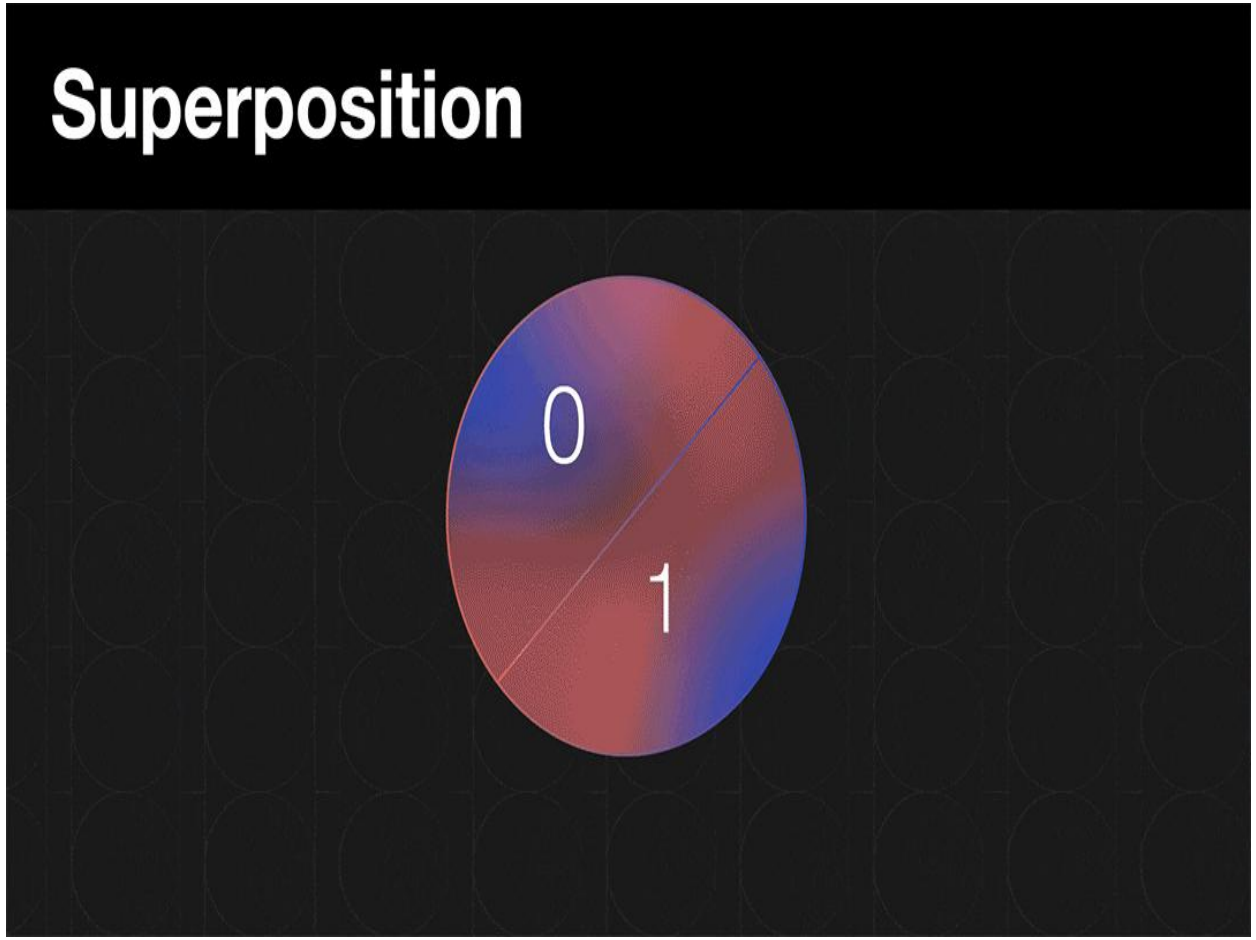
- To understand quantum computing, it is important to understand four key quantum mechanics principles:-
- **Superposition**
- **Entanglement**
- **Interference**
- **Measurement**



Four key principles of quantum mechanics

-Superposition

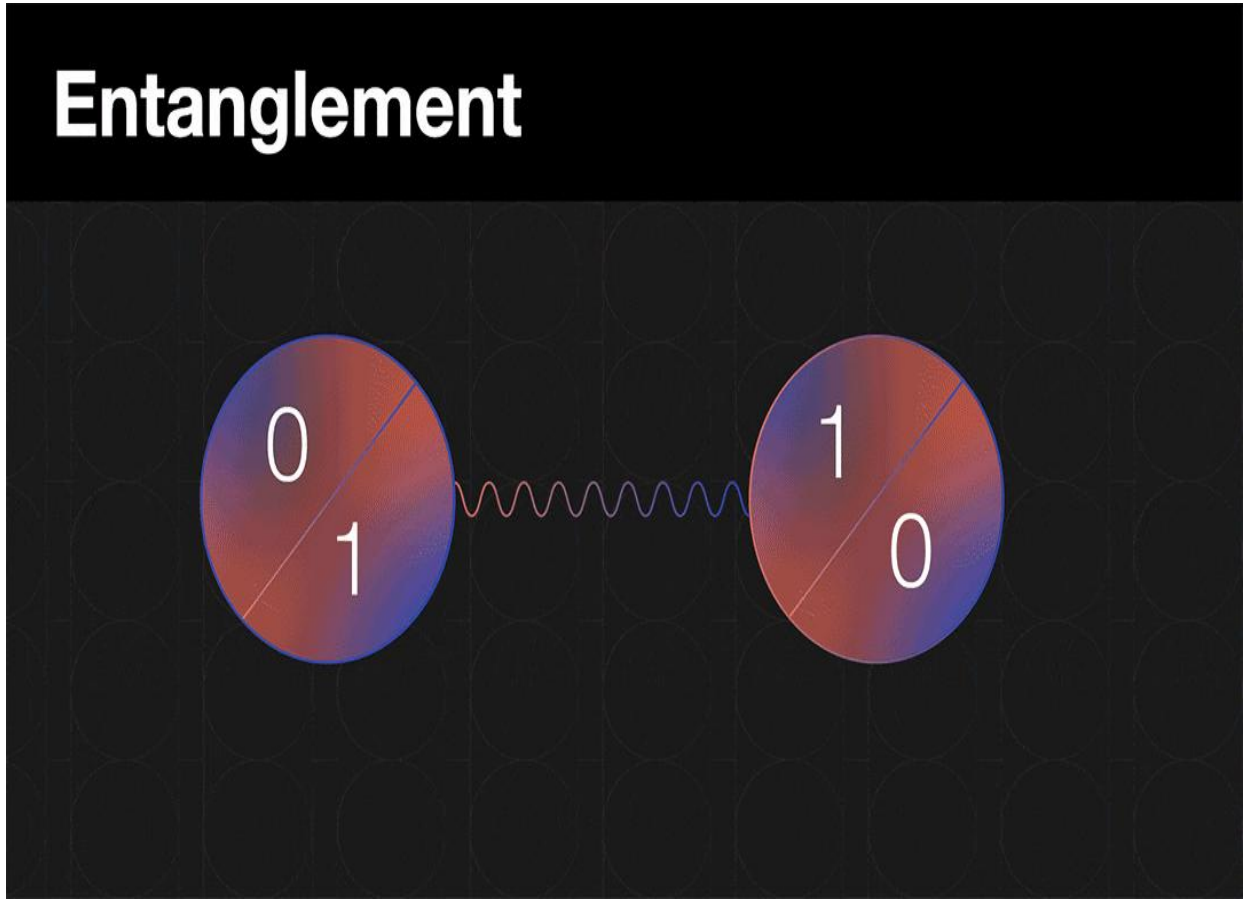
- A qubit itself isn't useful. But it can place the quantum information it holds into a state of superposition, which represents a combination of all possible configurations of the qubit.
- Groups of qubits in superposition can create complex, multidimensional computational spaces. Complex problems can be represented in new ways in these spaces.
- A qubit can exist in multiple states at once
- Not limited to only 0 or 1
- Enables parallel computation



Four key principles of quantum mechanics

-Entanglement

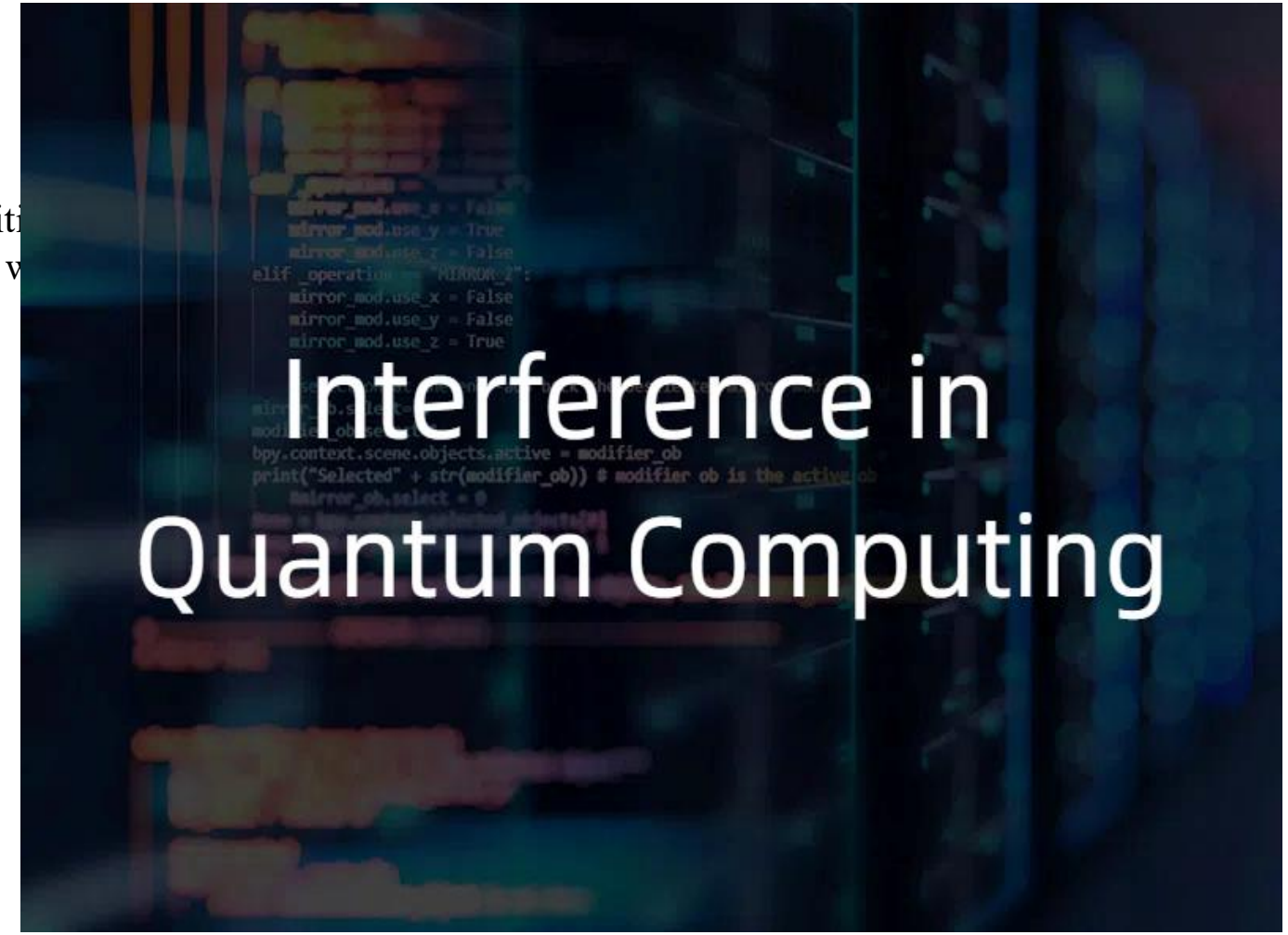
- Entanglement is the ability of qubits to correlate their state with other qubits
- Entangled systems are so intrinsically linked that when quantum processors measure a single entangled qubit, they can immediately determine information about other qubits in the entangled system.
- Qubits can be connected to each other
- Changing one affects the other instantly
- Important for quantum communication



Four key principles of quantum mechanics

-Interference

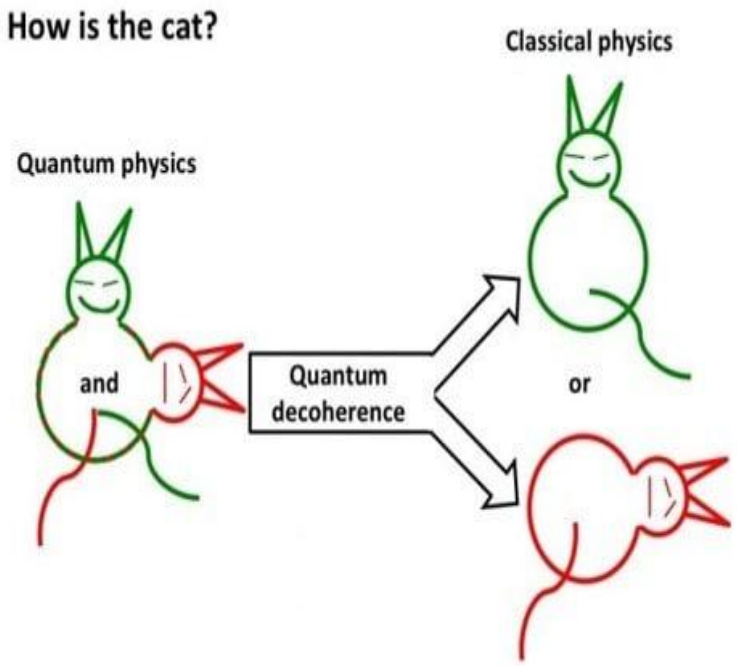
- Interference is the engine of quantum computing.
- When qubits are placed into a state of collective superpositi
  they structure information in a way that resembles waves, w
  amplitudes associated with each outcome.
- Quantum states can combine
- Helps amplify correct results
- Reduces incorrect outcomes



Four key principles of quantum mechanics

-Decoherence

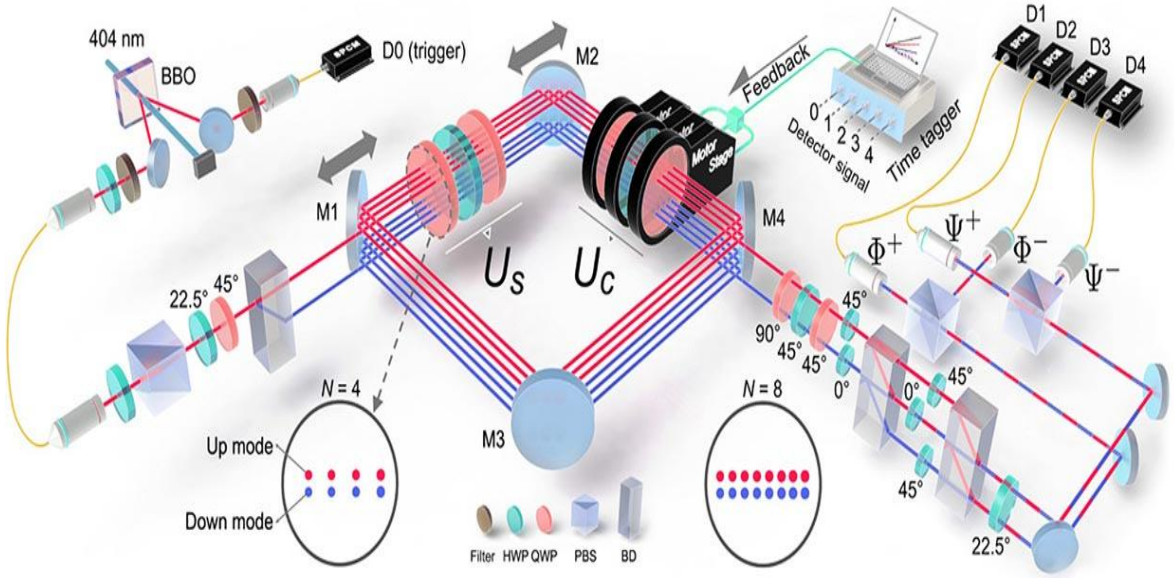
- Decoherence is the process in which a system in a quantum state collapses into a nonquantum state.
- It can be intentionally triggered by measuring a quantum system or by other environmental factors (sometimes these factors trigger it unintentionally). Generally speaking, quantum computing requires avoiding and minimizing decoherence.
- Observing a qubit changes its state
- It collapses into a definite value (0 or 1)
- Important in getting final results



Quantum decoherence explained using the thought experiment of Schrödinger's cat

How the principles work together

- To better understand quantum computing, consider that two surprising ideas are both true.
- The first is that objects that can be measured as having definite states qubits in superposition with defined probability amplitudes behave randomly.
- The second is that distant objects in this case, entangled qubit can still behave in ways that, though individually random, are tightly correlated.
- A computation on a quantum computer works by preparing a superposition of computational states. A quantum circuit, prepared by the user, uses operations to entangle qubits and generate interference patterns, as governed by a quantum algorithm.
- Many possible outcomes are canceled out through interference, while others are amplified. The amplified outcomes are the solutions to the computation



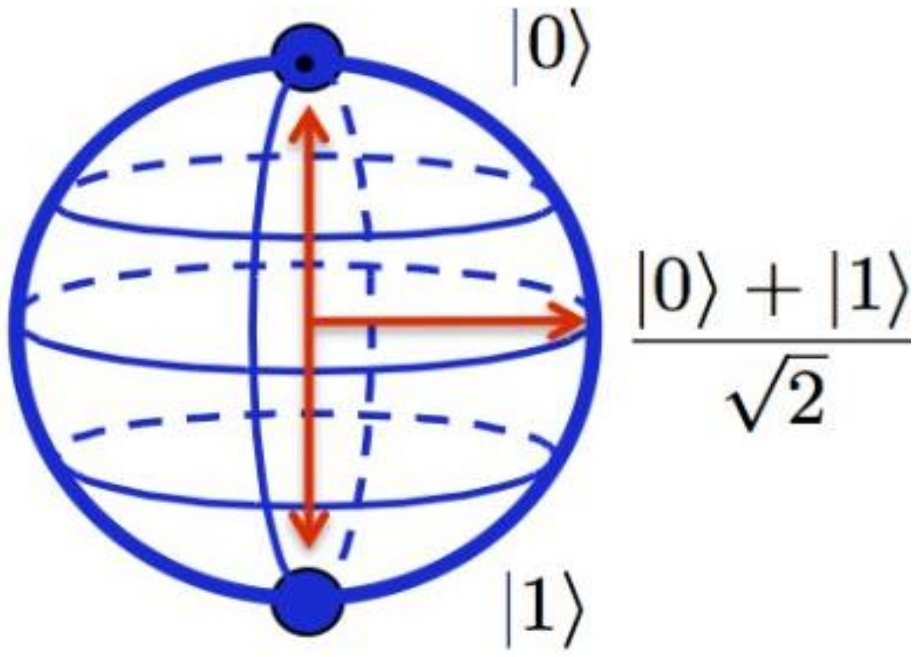
How do quantum computers work?

The primary difference between classical and quantum computers is that quantum computers use qubits instead of bits.

While quantum computing does use binary code, qubits process information differently from classical computers. But what are qubits and where do they come from?

What are qubits?

While classical computers rely on bits (zeros and ones) to store and process data, quantum computers process data differently by using quantum bits (qubits) in superposition.



Qubit

Types of qubits

Superconducting qubits: Made from superconducting materials operating at low temperatures, these qubits are favored for their speed in performing computations and fine-tuned control.

Trapped ion qubits: Trapped ion particles can also be used as qubits and are noted for long coherence times and high-fidelity measurements, but operate at a slower pace than superconducting qubits.

Quantum dots: Quantum dots are small semiconductors that capture a single electron and use it as a qubit, offering promising potential for scalability and compatibility with existing semiconductor technology.

Photons: Photons are individual light particles. They can be used to make qubits and send quantum information across long distances through optical fiber cables. They are being used in quantum communication and quantum cryptography.

Why are qubits useful?

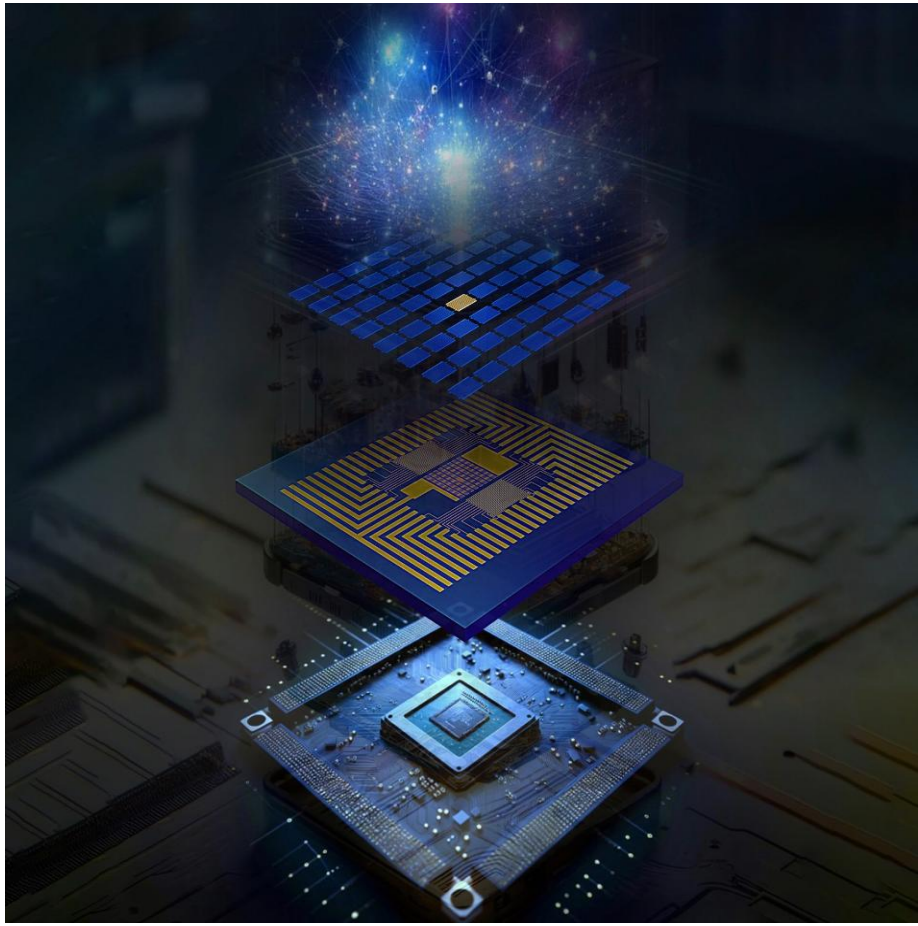
Computers that use quantum bits have certain advantages over computers that use classical bits. Because qubits can hold a superposition and exhibit interference, a quantum computer that uses qubits approaches problems in ways different from classical computers.

As a helpful analogy for understanding how quantum computers use qubits to solve complicated problems, imagine you are standing in the center of a complicated maze.

Quantum computing components

- Quantum computers use specialized hardware systems
- These systems operate at ultracold temperatures
- Additional room-temperature electronics control operations
- The complete system is large and complex
- Many quantum computers are accessed through cloud computing

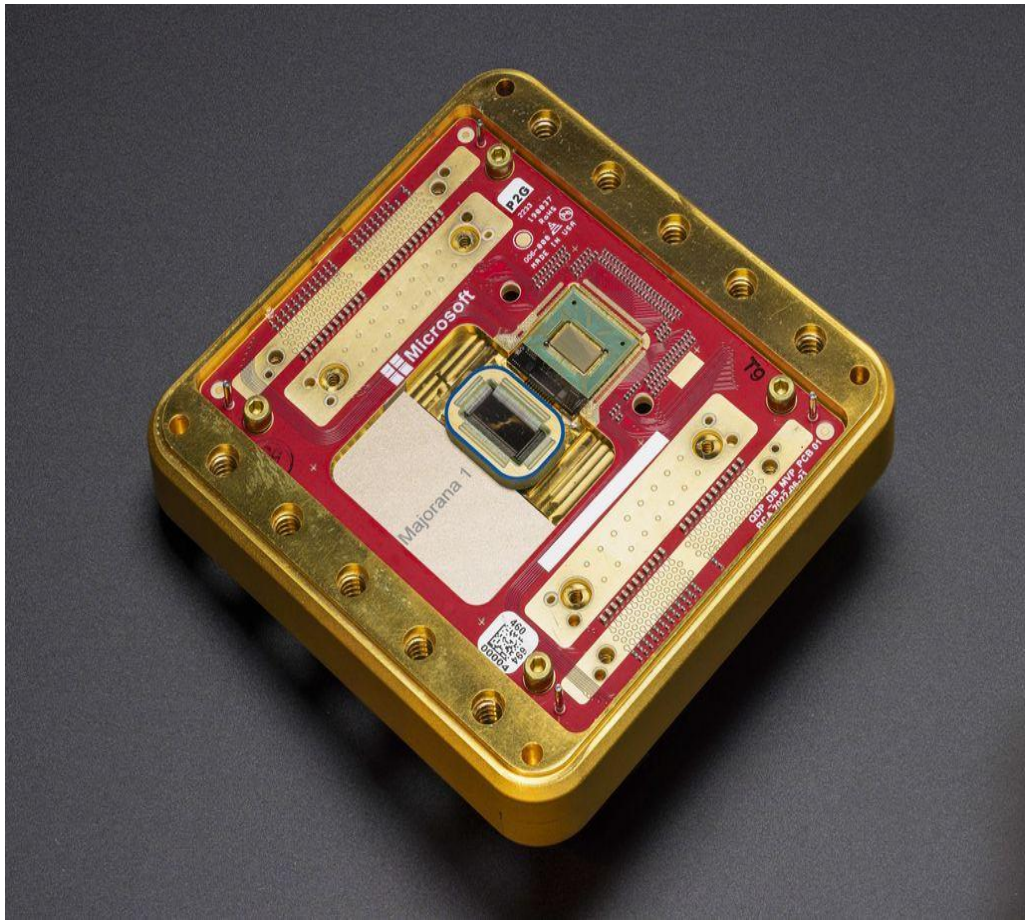
- **Quantum processors**
- **Superconductors**
- **Control**
- **Quantum software**



Quantum computing components

-Quantum processors

- Quantum processors are the core of a quantum computer
- Made up of qubits arranged for communication
- Also called the quantum data plane
- Act as the brain of the system
- Includes quantum chips, control electronics, and classical hardware



Quantum computing components

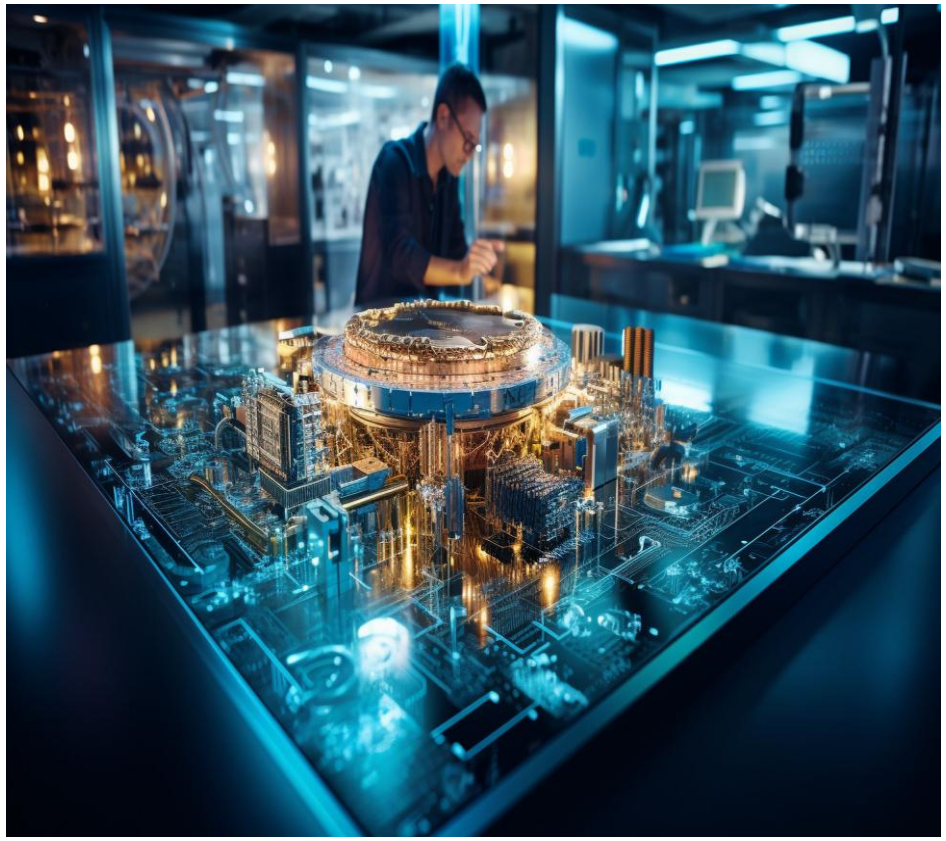
-Superconductors

- Quantum processors require extremely low temperatures
- About near absolute zero to reduce noise and errors
- At these temperatures, materials become superconductors
- Electrons move without resistance
- Enables stable quantum operations

Quantum computing components

-Control

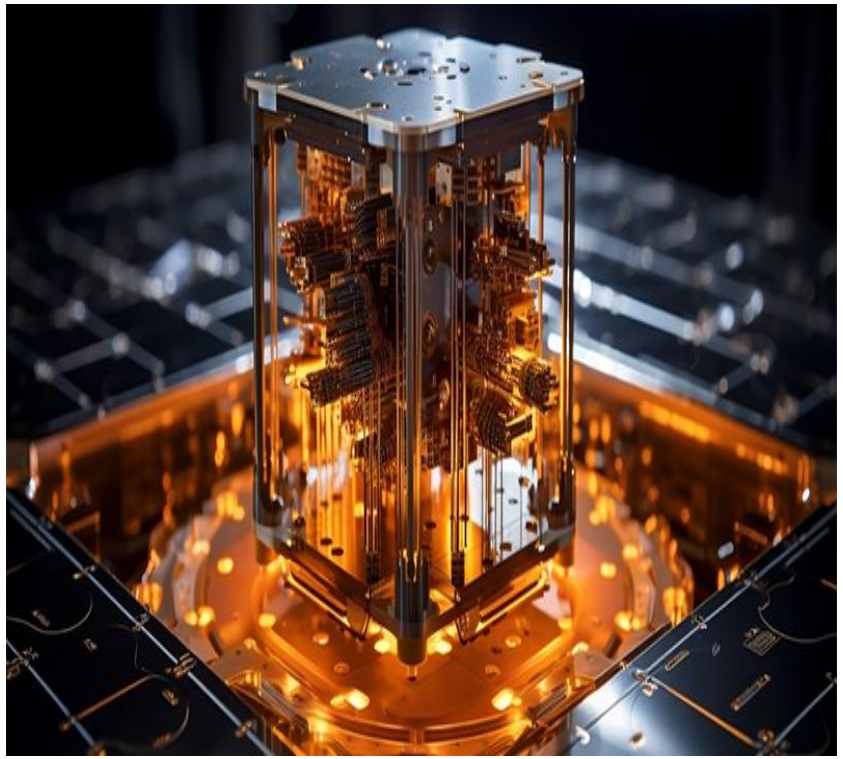
- Quantum computers use circuits with capacitors and junctions
- Controlled using microwave signals (photons)
- These signals:
- Control qubit behavior
- Store information
- Read quantum data



Quantum computing components

-Quantum software

- Hardware alone is not enough for quantum computing
- Software enables execution of quantum algorithms
- Required for writing, optimizing, and running programs
- Essential for achieving quantum advantage



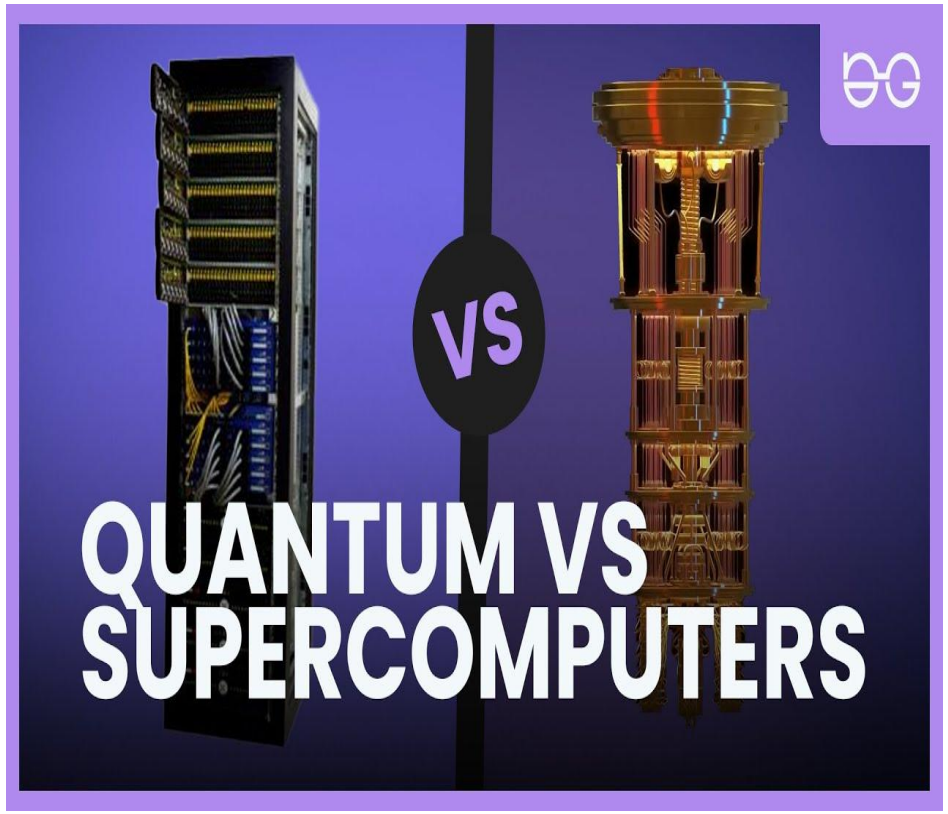
Classical computing versus quantum computing

Classical computer	Quantum computer
From antiquated punch-card adders to modern supercomputers, traditional (or classical) computers essentially function in the same way. These machines generally perform calculations sequentially, storing data by using binary bits of information. Each bit represents either a 0 or 1.	Quantum computers, like classical computers, are problem-solving machines. But instead of bits, quantum computing uses qubits. Qubits are used to process data like traditional bits; however, by harnessing quantum phenomena, qubits have access to more complex mathematics for a different type of computation. This principle is due to quantum mechanical concepts known as superposition and interference, which were discussed earlier.
Used by common, multipurpose computers and devices.	Used by specialized quantum mechanics-based quantum computing hardware.
Processes information in bits with a discrete number of possible states, 0 or 1.	Processes information in qubits as 0, 1 or a superposition of 0 and 1.

When is quantum computing superior?

- Classical computers are best for most everyday tasks
- Quantum computing is useful for very complex problems
- Complex problems involve many variables interacting together

  Example: simulating how molecules behave
- Classical computers:
  - Solve problems step-by-step
  - Struggle with very large and complex calculations
- Quantum computers:
  - Use a different approach to solve problems
  - Can model complex systems directly
- More efficient for problems like chemical simulations
- Key idea:
- **Classical** → calculate step-by-step
- **Quantum** → mimic the system directly

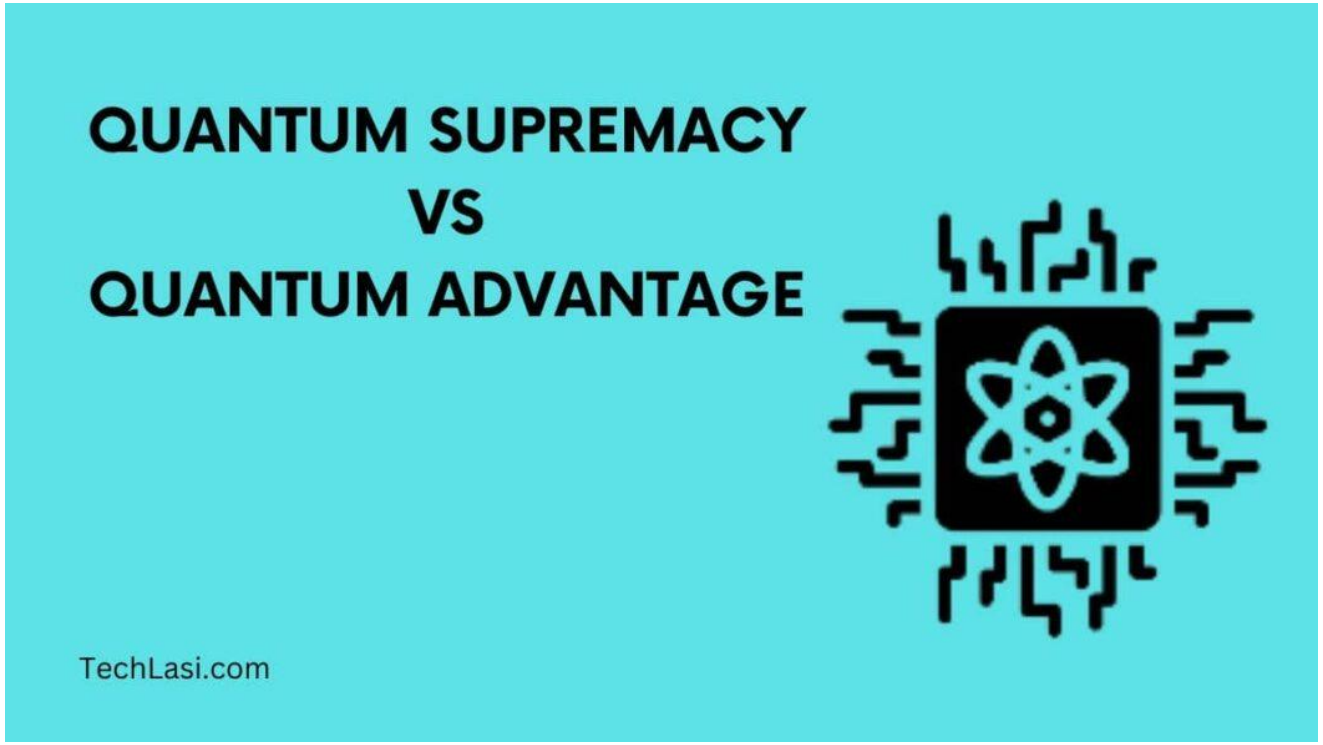


Quantum computing use cases

- Quantum computing has real-world applications across many industries
- In 1994, Peter Shor showed that quantum computers can break modern cryptography
- This proved quantum computing can solve important real problems
- Many industries are exploring quantum computing:
  - **Engineering**
  - **Finance**
  - **Logistics and shipping**
- Quantum computing is useful for:
  - **Solving complex problems with many variables**
  - **Simulating molecular behavior**
  - **Managing energy systems**
  - **Financial modeling**
- Key industries using quantum computing:
  - **Pharmaceuticals** → faster drug discovery
  - **Chemistry** → better materials and cleaner processes
  - **Machine Learning** → improved data analysis

Quantum utility vs Quantum advantage

- Quantum computing is still developing but improving rapidly
- Scientists are working to make quantum computers more useful
- and powerful
- Two important concepts:
  - **Quantum utility**
  - **Quantum advantage**
- **Quantum utility**
  - When quantum computers solve problems that classical computers cannot solve directly
  - Provides reliable and accurate results
  - Better than traditional approximation methods
  - First demonstrated by IBM in 2023

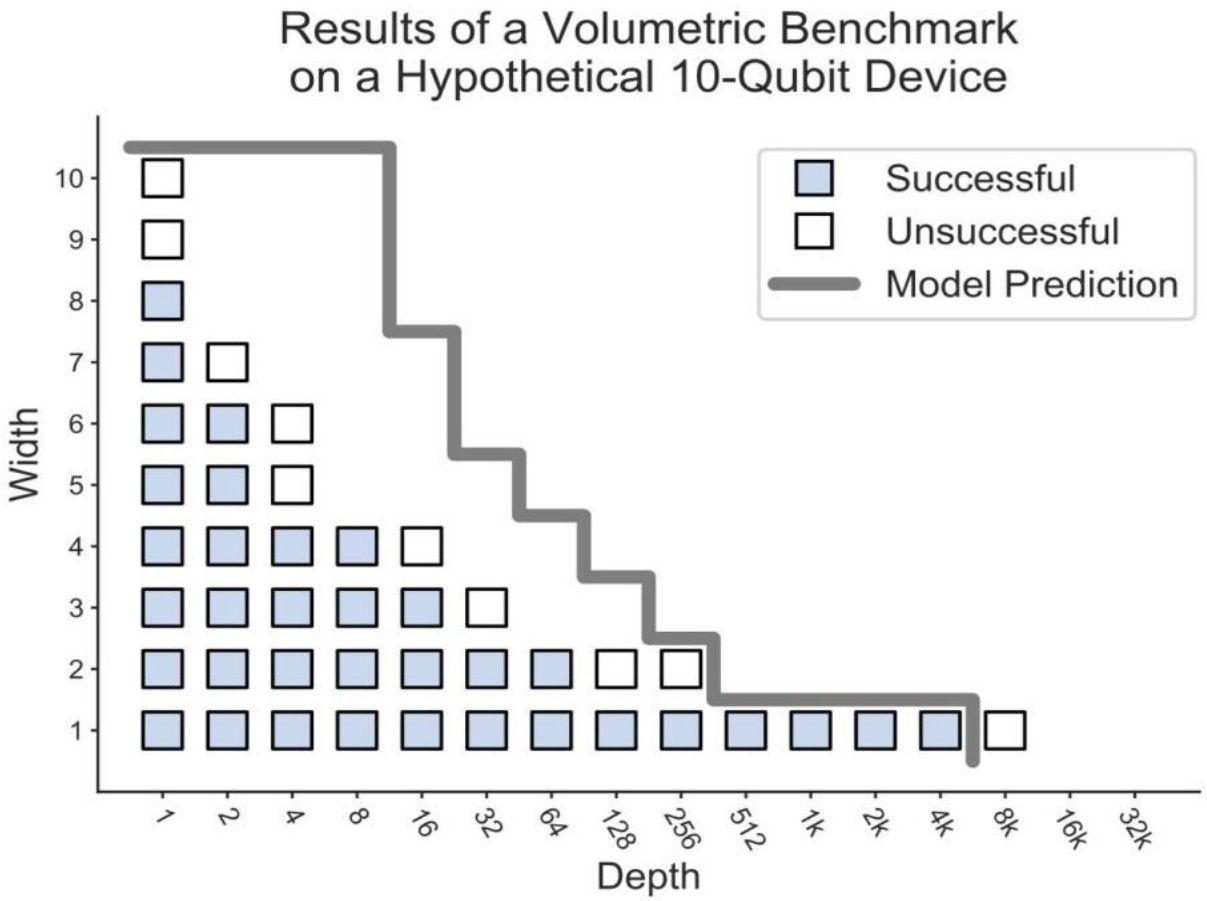


- **Quantum advantage**
  - When quantum computers perform better than all classical methods
  - Can be faster, cheaper, or more efficient
  - Provides real practical benefits
  - Expected to be achieved in the near future

Quantum Benchmarks

- Quantum benchmarks are used to measure the performance of quantum computers
- Quantum computing is useful for some problems, but not always faster than classical methods
- This is different from quantum advantage

- **Layer fidelity**
  - Measures how well a quantum processor runs circuits
  - Shows performance of qubits, gates, and errors
  - Helps evaluate overall system quality
- **CLOPS (Circuit Layer Operations Per Second)**
  - Measures speed of quantum processing
  - Shows how fast quantum circuits are executed
  - Includes both quantum and classical system performance.
- **Circuit depth**
  - Measures how many steps a quantum circuit can run
  - Higher depth → more complex problems can be solved
  - Limited by qubit stability (decoherence)



Quantum challenges and improvements

- Quantum computing is advancing rapidly with support from major companies
- Many organizations are working to improve quantum systems
- Quantum computers are now real but still need improvement

- **Scaling quantum processors**
  - Current systems support limited qubits
  - Future goal: more powerful and larger systems
  - IBM plans 200 logical qubits and 100 million gates by 2029
  - Long-term goal: 2,000 qubits and 1 billion gates

- **Scaling hardware**
  - Requires extremely low temperatures
  - High cost and large infrastructure
  - Need to reduce size and energy use
  - Work is ongoing to improve hardware efficiency

Quantum challenges and improvements

- **Quantum error correction**
  - Qubits are unstable and error-prone
  - Errors must be corrected for accurate results
  - Decoherence affects performance
  - New methods are improving efficiency

- **Quantum algorithm development**
  - Need better algorithms for real-world problems
  - Required to achieve quantum advantage
  - Focus on solving problems better than classical methods

- **Quantum software**
  - Strong software needed to run quantum programs
  - Tools like Qiskit help development

References

- https://www.ibm.com/think/topics/quantum-computing

Thank You